Testing model-dependent behaviour

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Testing model-dependent behaviour before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Testing pure components

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Authentication and authorisation

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Architecture and data flow

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence.

Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.

Practise with fading help

> Complete the supported Testing model-dependent behaviour trace and record the first step you can explain without help.

Use W2: Trace the worked case for Testing model-dependent behaviour. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Testing model-dependent behaviour explanation against the supplied facts before discussing it.

Use the worked case for Testing model-dependent behaviour. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Testing model-dependent behaviour facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Testing model-dependent behaviour facts and keep your reasoning.

Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Testing model-dependent behaviour facts and keep your reasoning.

A report omits this qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Testing model-dependent behaviour facts and keep your reasoning.

Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Testing model-dependent behaviour facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Testing model-dependent behaviour to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled testing model-dependent behaviour **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing model-dependent behaviour.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

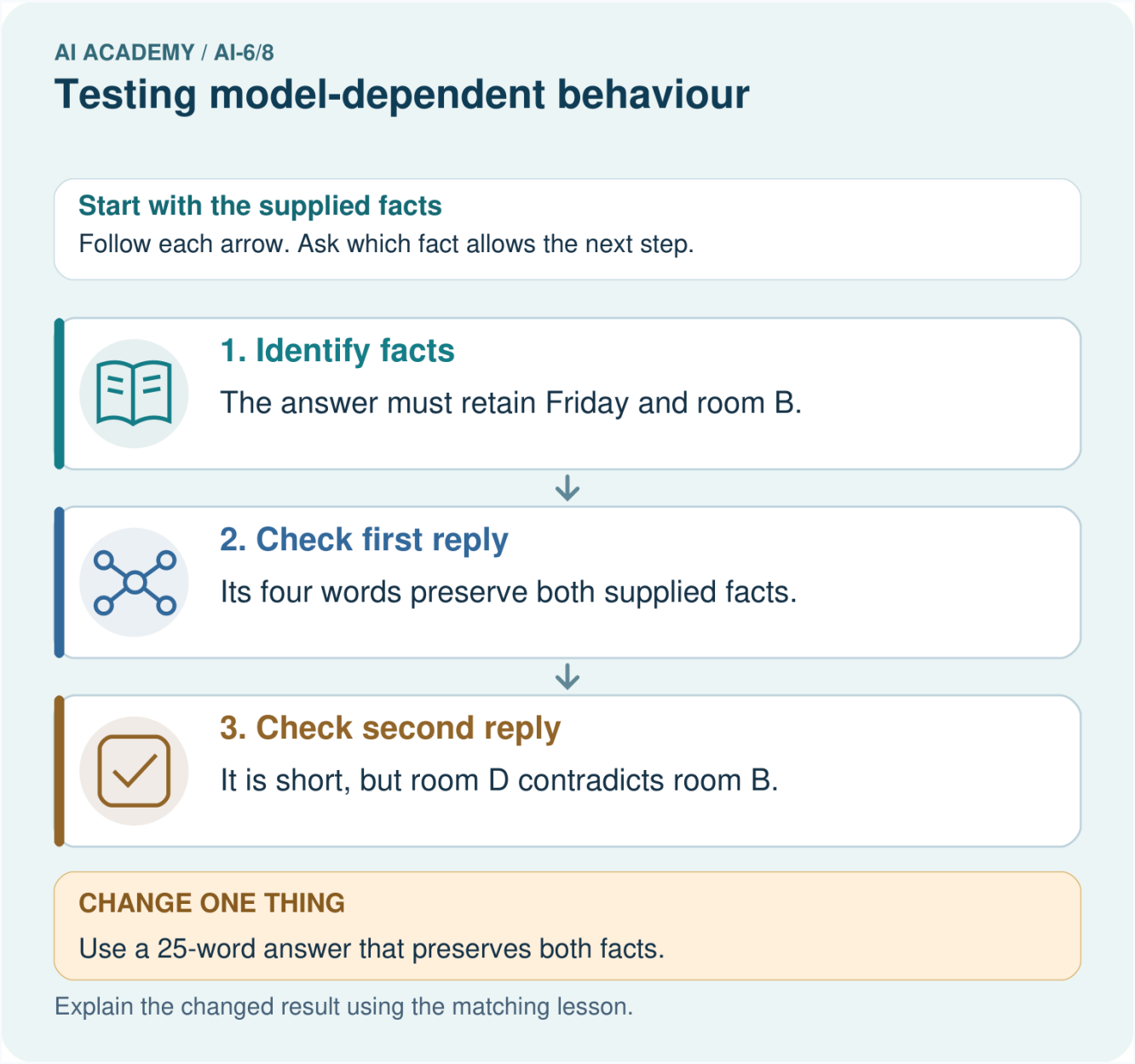
Pictorial case practice: Testing model-dependent behaviour

> Judge two summaries with a rubric. A summariser must preserve two supplied facts and use

> at most 20 words. Explain the labelled step before checking the changed condition.

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".



Follow the labelled picture with the matching explanation. Open the vector diagram

Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? It is short, but room D contradicts room B.

1. Check second reply

2. Check first reply

3. Identify facts

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check second reply It is short, but room D contradicts room B. A behaviour contract can accept different wording while requiring the same facts.

Short, fluent text may still fail factual support.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use a 25-word answer that preserves both facts. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. It passes support but fails the length requirement.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It passes support but fails the length requirement. Every mandatory criterion must pass; facts do not erase a length violation.

Repair a missing building block

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the **mechanism** and terminology of authentication and authorisation. Short practice: Explain why Jo’s valid volunteer login does not permit editing C.

Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

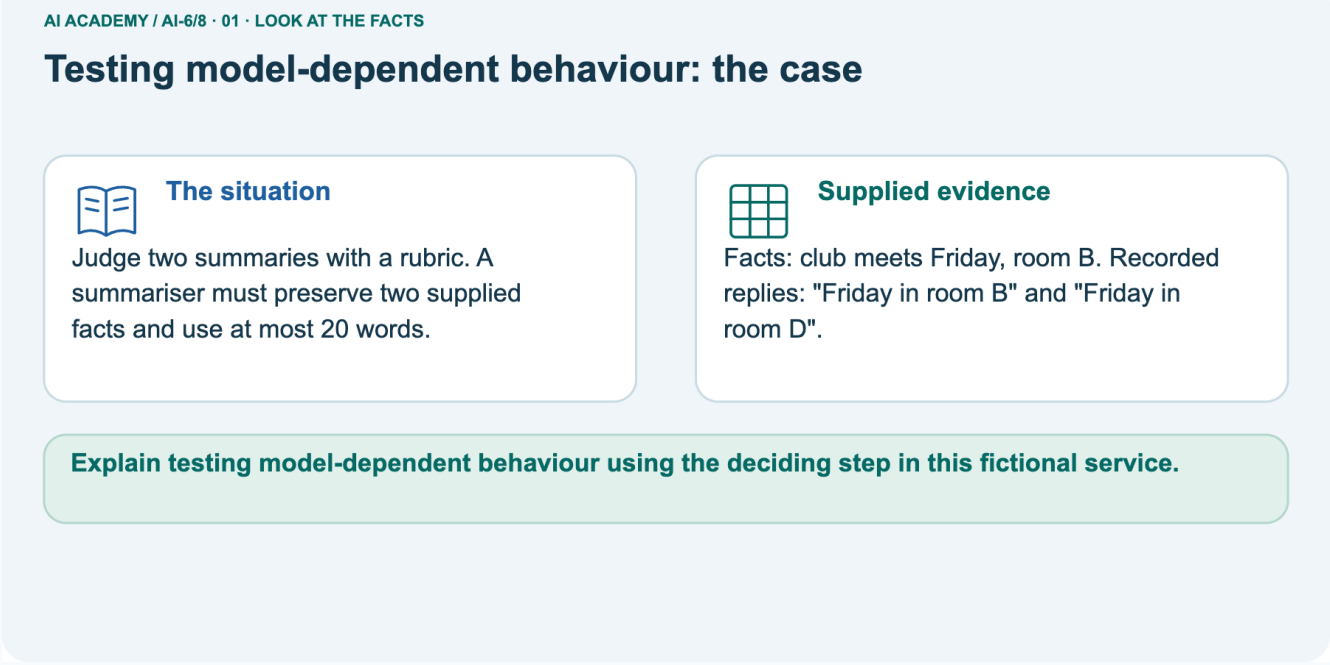
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Testing model-dependent behaviour picture from its labels, then record a relevant limit.

01. Testing model-dependent behaviour: the case



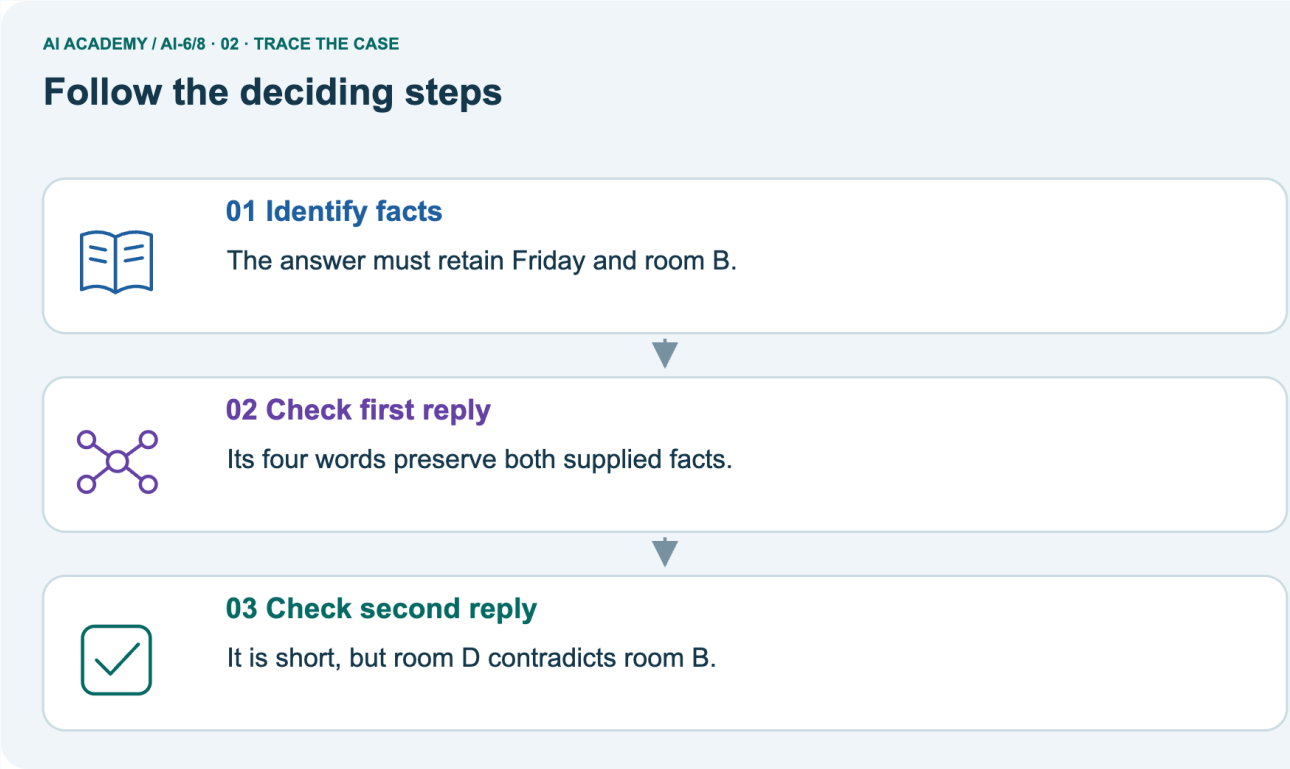
Follow the labelled picture with the matching explanation. Open the vector diagram

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words. Supplied facts: Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



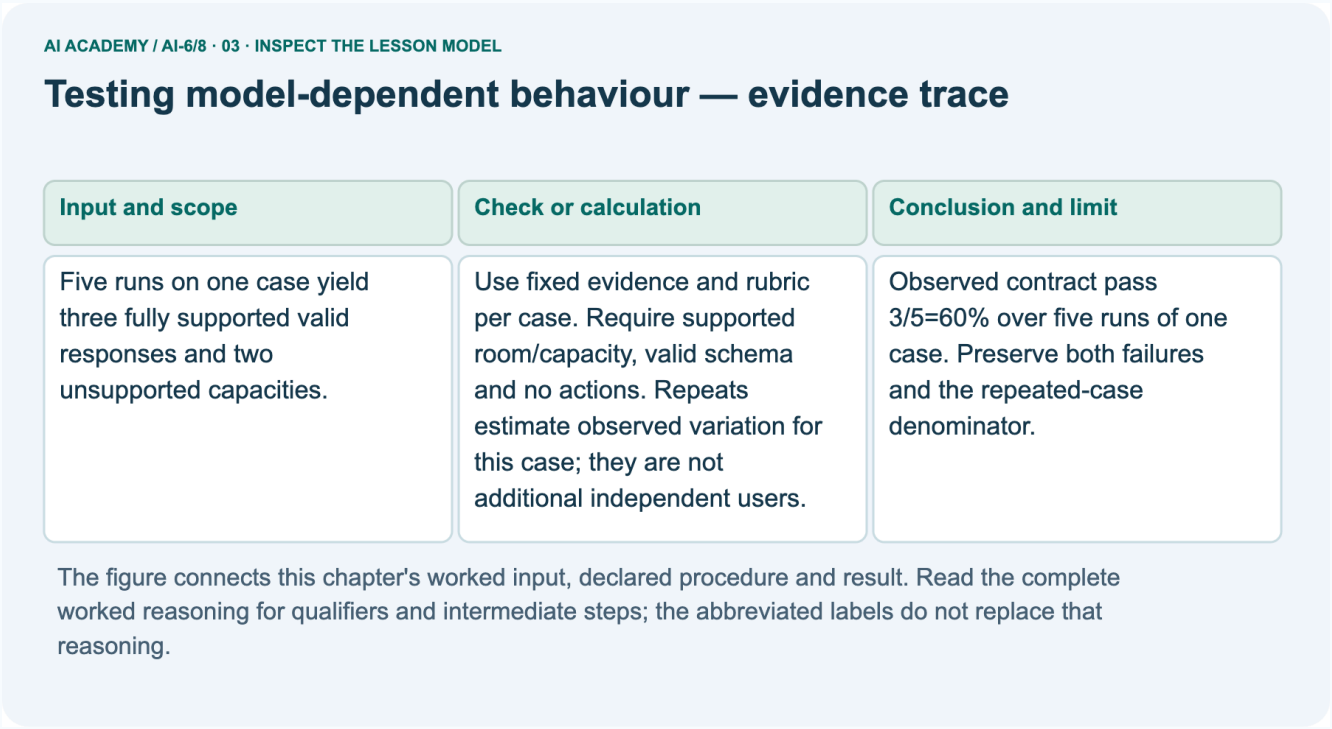
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify facts: The answer must retain Friday and room B. → Check first reply: Its four words preserve both supplied facts. → Check second reply: It is short, but room D contradicts room B.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Testing model-dependent behaviour — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing model-dependent behaviour — **evidence** trace. Five runs on one case yield three fully supported valid responses and two unsupported capacities. | Use fixed **evidence** and rubric per case.

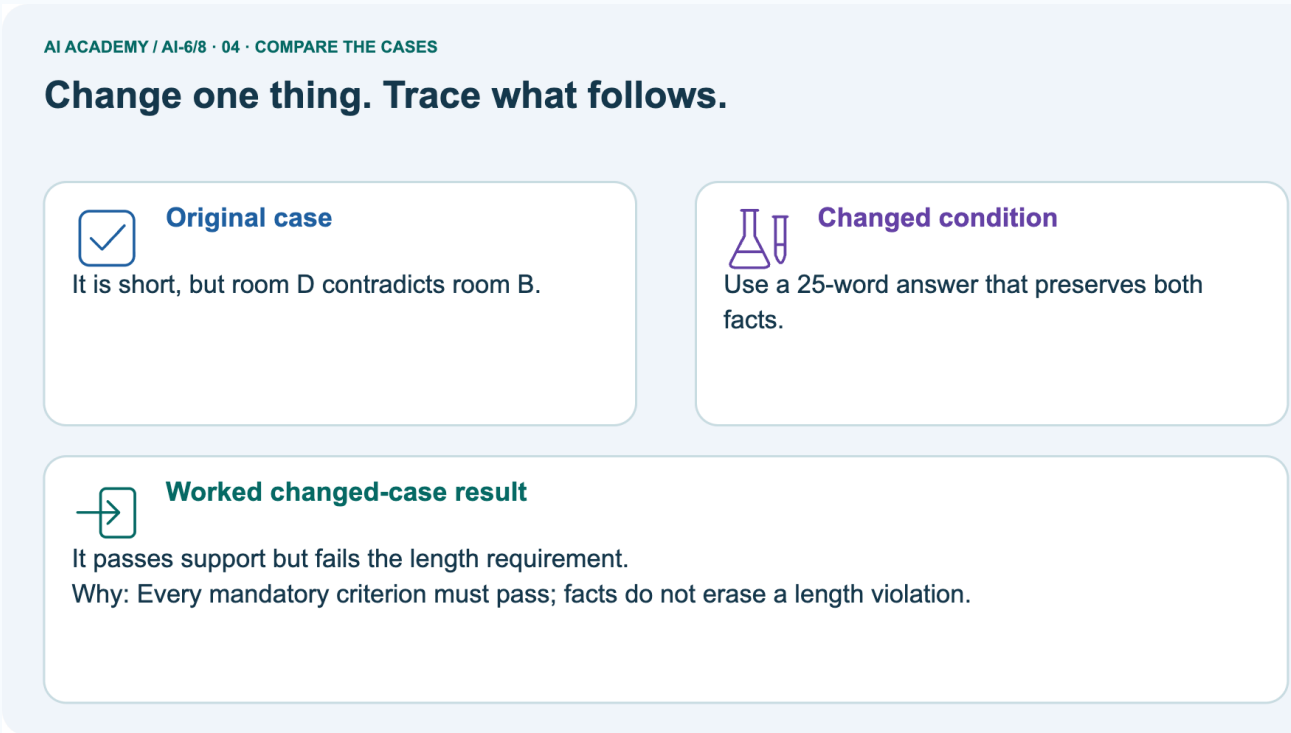
Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users. | Observed contract pass 3/5=**60%** over five runs of one case.

Preserve both failures and the repeated-case denominator.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

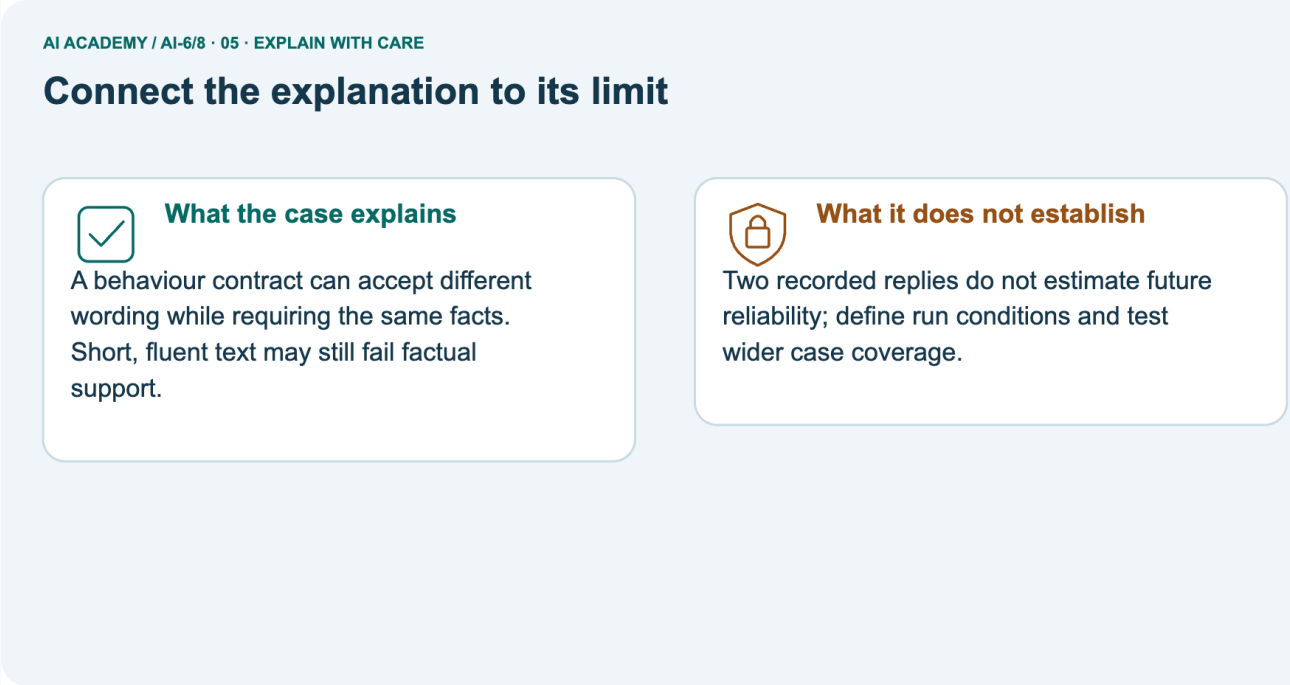
Original case: It is short, but room D contradicts room B. Changed condition: Use a 25-word answer that preserves both facts. Expected result: It passes support but fails the length requirement.

Reason: Every mandatory criterion must pass; facts do not erase a length violation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

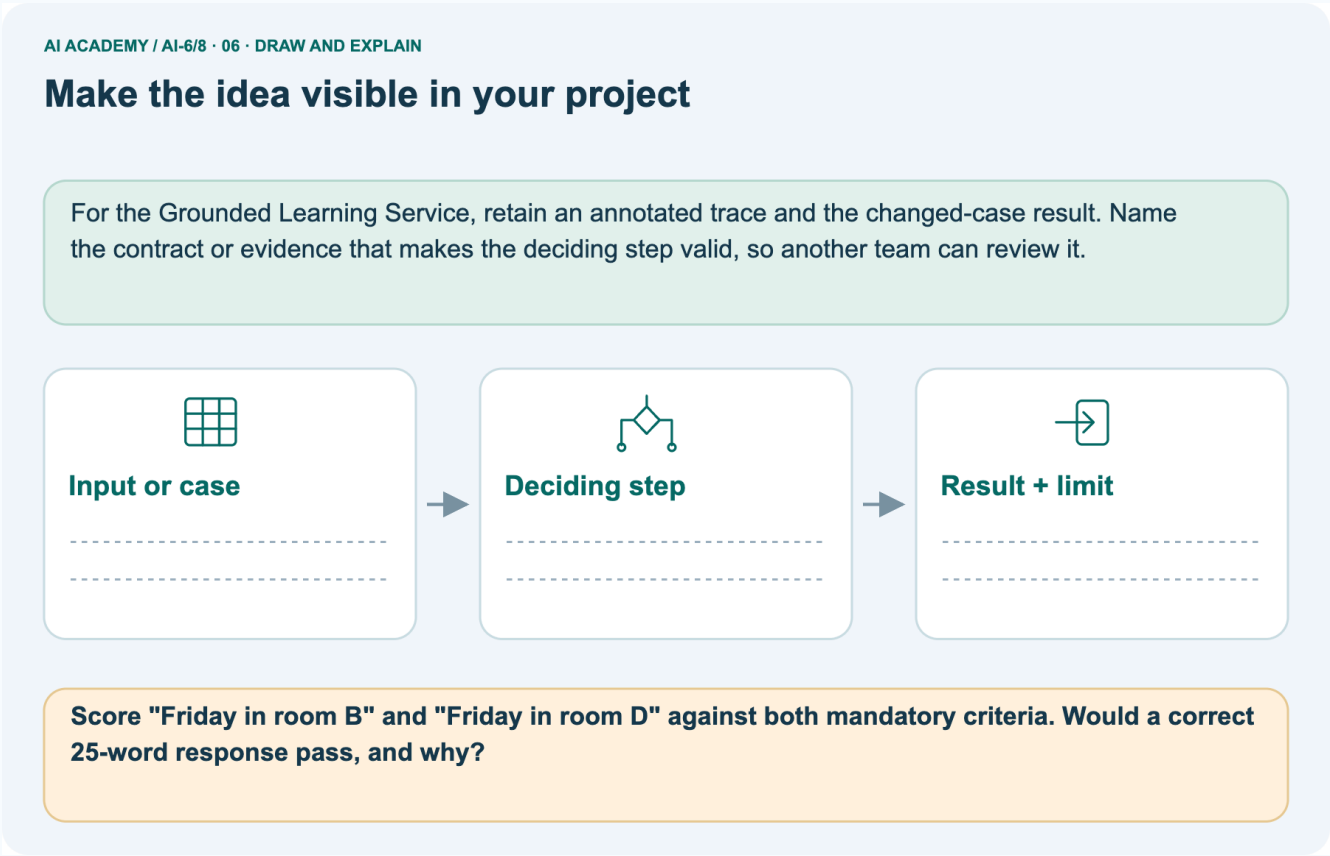
A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

Limit: Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.